

EpiCoCo: *De Novo* Epitope Generation via MHC-Context Co-Modeling and Contrastive Affinity Guidance

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Abstract

The *de novo* generation of high-affinity epitopes tailored to specific major histocompatibility complex (MHC) proteins is a pivotal challenge in computational immunotherapy. However, current methods struggle to effectively integrate the MHC context into the generation process, and often fail to guarantee high binding affinity due to the neglect of discriminative signals from non-binders. To bridge these gaps, we present **EpiCoCo**, a probabilistic framework for **Epitope** generation via MHC-context **Co**-modeling and **Contrastive** affinity learning. EpiCoCo treats the pMHC complex as a dynamic, co-adaptive system by operating on the joint E(3) graph. In addition, we introduce Contrastive Affinity Guidance (CAG), an inference mechanism that leverages the gradient difference between learned high- and low-affinity distributions. CAG actively drives the generation trajectory towards high-affinity manifolds while utilizing repulsive signals to filter out candidates with poor binding potential. Extensive evaluations demonstrate that EpiCoCo achieves a mean binding free energy of -45.20 REU, a 23% improvement over the state-of-the-art, while maintaining high structural plausibility. The results validate that context co-modeling and negative-informed guidance are essential for generating valid, high-potency immunotherapeutics. Our code is publicly available at <https://github.com/StevenLuan904/EpiCoCo>.

1. Introduction

Peptide-based therapeutics, such as cancer neoantigen vaccines, represent a programmable paradigm for precision

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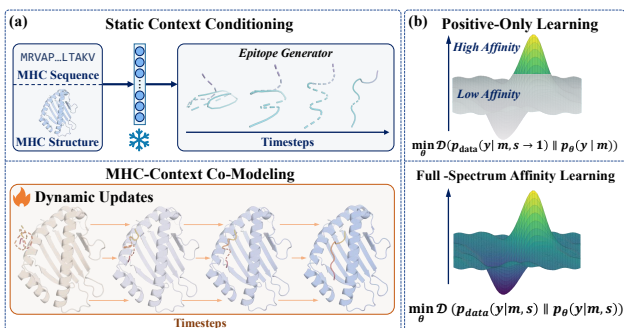


Figure 1. Conceptual comparison between conventional paradigms and EpiCoCo. (a) **Generative Modeling:** Unlike static conditioning methods that treat the MHC as a frozen template (top), **MHC-Context Co-Modeling** (bottom) allows the MHC to co-adapt with the epitope. (b) **Learning Objective:** While standard approaches rely on *Positive-Only Learning* that neglects the decision boundary (Top), **Full-Spectrum Affinity Learning** (Bottom) leverages continuous affinity labels to map the complete energy landscape and utilize abundant negative data to navigate towards high-affinity modes robustly.

medicine. Their efficacy relies fundamentally on major histocompatibility complex (MHC) molecules presenting specific peptide sequences (epitopes) to trigger an immune response (Neefjes et al., 2011; Berkers et al., 2015). This frames vaccine development as a precise matching problem: generating peptides that bind exclusively to patient-specific MHC alleles (Gigoux et al., 2022; Shah et al., 2025). The difficulty lies in the fact that peptide-MHC (pMHC) interactions are not static. They define a deeply coupled system where binding affinity is governed by structural adaptability rather than sequence identity alone. Effective generation requires resolving the induced-fit mechanism, where the epitope and MHC groove dynamically adjust to form essential anchors (Wieczorek et al., 2017a). Hence, an optimal strategy must jointly generate the sequence and geometry of the complex to resolve intrinsic physicochemical coupling.

However, current generative paradigms fall short of this requirement by treating the receptor context as a static condition embedding, thereby neglecting the requisite structural co-adaptation between the peptide and the receptor. Most approaches rely on either 1D sequence (Chen et al., 2025a; Bhat et al., 2025) or rigid structural embeddings (Kong et al., 2024; Lin et al., 2024; Li et al., 2024; Wang et al., 2024a;

Kong et al., 2025), thereby treating the MHC receptor as a frozen template. For instance, discrete tokenization strategies DPLM-2 (Wang et al., 2025) inherently fragment the continuous E(3) manifold, disrupting the continuous spatial dependencies required to model flexible docking. Similarly, continuous frameworks like PepFlow (Li et al., 2024) and PepMimic (Kong et al., 2025) typically encode the MHC as a rigid boundary condition. This oversimplifies the binding physics by neglecting physicochemical complementarity, where the MHC groove undergoes essential conformational changes upon binding (Wieczorek et al., 2017a). For detailed analysis on related works, see Appendix Section A.

Beyond representation, a critical bottleneck remains in training objectives. Standard generative models for peptide design predominantly rely on *Positive-Only Learning*. By maximizing likelihood exclusively on binders, these methods capture the high-affinity manifold but lack the gradient signals necessary to suppress the probability of non-binding regions. This results in undefined decision boundaries and high false-positive rates. Compounding this issue, the quantization of continuous affinity into binary labels collapses the intricate potential energy surface into a coarse step function, discarding the fine-grained ranking information essential for specificity.

To address these challenges, we present **EpiCoCo**, a structure-sequence co-modeling framework designed for high-specificity epitope generation. Operating on the joint E(3) manifold of the entire pMHC complex, EpiCoCo treats the MHC not as a fixed template, but as a flexible scaffold that actively adapts during generation. This enables our model to characterize the dynamic geometry that governs binding specificity. Furthermore, EpiCoCo integrates continuous affinity labels as soft prompts, enabling the network to map the underlying energy landscape and distinguish high-affinity modes from low-affinity noise. Building on this, we introduce **Contrastive Affinity Guidance (CAG)**, an inference-time mechanism that exploits the learned landscape. CAG computes the gradient difference between positive and negative distributions to construct a guidance signal that propels generation towards a sharpened high-affinity manifold. This process actively filters out suboptimal conformations, resolving an inherent limitation of positive-only learning. Our contributions are summarized as follows:

- We propose EpiCoCo, a probabilistic framework that co-generates epitope sequences and structures within the pMHC complex. This **flexible MHC-context co-modeling** allows for induced-fit adjustments to ensure both structural validity and binding specificity.
- We propose **Full-Spectrum Affinity Learning** strategies. By integrating affinity scores as soft prompts during training and employing CAG during inference,

our approach leverages the energy landscape to effectively drive generation towards high-affinity manifolds.

- EpiCoCo outperforms state-of-the-art methods in generative quality, achieving a mean binding free energy of -45.20 REU, while maintaining the highest structural confidence (average 0.89 pLDDT). Furthermore, EpiCoCo distinctively supports **negative-informed generation**, discovering high-affinity binders even when trained exclusively on negative (non-binder) data. This capability highlights the robustness and underlying generalization power of our contrastive learning paradigm.

2. Methodology

2.1. Problem Formulation

Let $\mathcal{A}$ be the set of standard amino acid residues. We define a peptide as a sequence $y = (y_1, \dots, y_L) \in \mathcal{A}^*$, with L indicating the variable length of the chain and $\mathcal{A}^*$ denoting the space of all finite sequences constructed from $\mathcal{A}$. Our goal is to model the continuous distribution of peptide sequences across the affinity spectrum. We formulate this as **Full-Spectrum Affinity Learning**. The objective is to approximate the conditional distribution $p_{\text{data}}(y | m, s)$, where m denotes the target MHC protein, and $s \in [0, 1]$ is an affinity scalar. We achieve this by minimizing the distance between the parameterized distribution and the conditional distribution:

$$\min_{\theta} \mathcal{D}(p_{\text{data}}(y | m, s) \parallel p_{\theta}(y | m, s)), \quad (1)$$

where $\mathcal{D}$ denotes a probabilistic divergence metric. By conditioning the generative process on s , the model effectively learns a tunable affinity landscape, allowing us to traverse the manifold from non-binding decoys ($s \rightarrow 0$) to high-affinity binders ($s \rightarrow 1$) during inference.

2.2. Overview of EpiCoCo framework

EpiCoCo performs structural co-generation via an E(3)-equivariant diffusion model. The framework incorporates **Affinity Soft Prompting (ASP)** to enable fine-grained learning of affinity, and **Contrastive Affinity Guidance (CAG)** to repel from non-binding modes while guiding generation towards high-affinity states. Implementation details are provided in Appendix Section B.

2.3. Continuous pMHC Diffusion

We implement the co-generation process via a variance-preserving diffusion framework operating on the joint system state $\mathbf{Z}_t = [\mathbf{X}_t, \mathbf{H}_t] \in \mathbb{R}^{N \times (3+d)}$. Here, $\mathbf{X}_t \in \mathbb{R}^{N \times 3}$ denotes the Cartesian coordinates of residues. $\mathbf{H}_t \in \mathbb{R}^{N \times d}$ represents the amino acid features derived from the diffusion process applied to the initial one-hot encoding, where

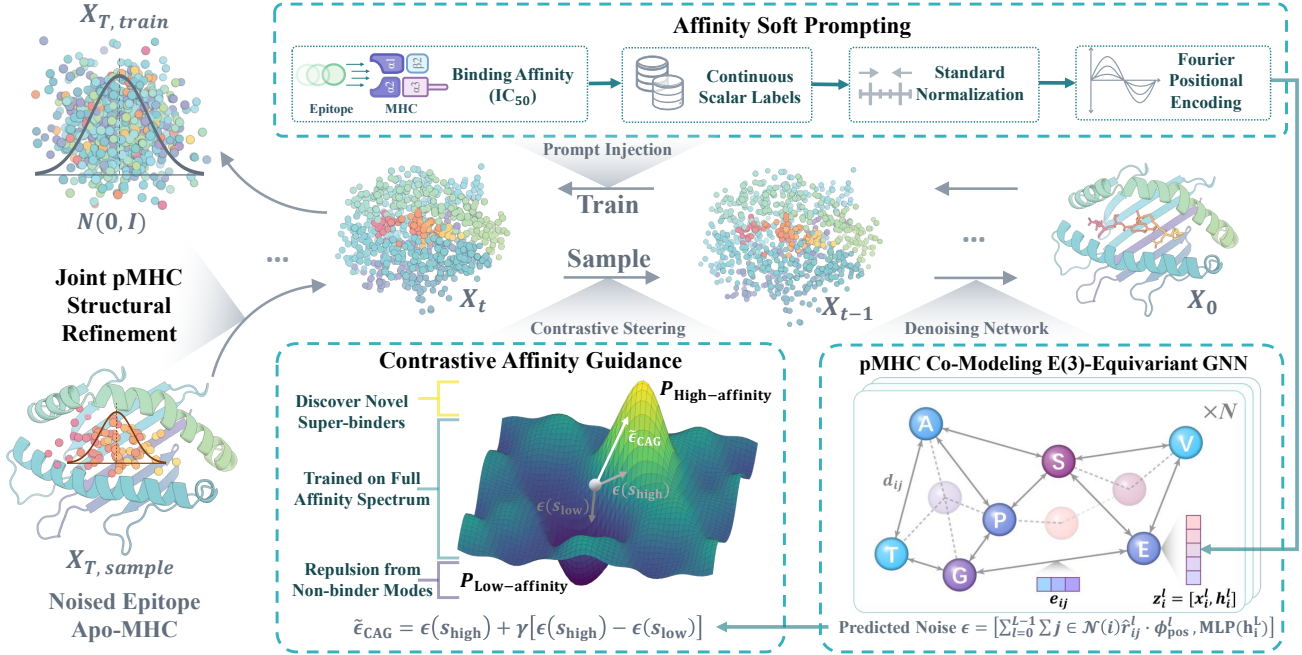


Figure 2. Overview of the EpiCoCo framework. The model treats the pMHC complex as a dynamic, co-adaptive system via a joint E(3)-equivariant graph diffusion process conditioned on binding affinity. During training, continuous scalar affinity labels are encoded via **Affinity Soft Prompting (ASP)** (top) to modulate the noise prediction network. During inference, the Apo-MHC (epitope-free state protein) serves as an initial scaffold, then dynamically co-adapts alongside the emerging epitope (bottom left). A novel **Contrastive Affinity Guidance (CAG)** mechanism (bottom center) exploits the contrastive gradients between high- and low-affinity conditions to steer the generative trajectory towards discovering potent binders.

d corresponds to the number of amino acid categories (e.g., 20) and N is the total number of residues.

To capture the underlying interactions between the epitope and the MHC, we treat the pMHC complex as one entity. During training, the forward process $q(\mathbf{Z}_t|\mathbf{Z}_0)$ injects Gaussian noise into the entire complex according to $\tilde{\mathbf{Z}}_t = \sqrt{\alpha_t}\mathbf{Z}_0 + \sqrt{1 - \alpha_t}\epsilon$. By modeling them together, the network learns to coordinate the geometry and sequence specifically for high-affinity binding. Implementation details are provided in Appendix Section B.

The inference process employs a scaffold-initialized strategy. We initialize the reverse trajectory at $t = T$ by composing a predicted MHC structure with random noise for the epitope:

$$\mathbf{Z}_T = \mathbf{M} \odot \mathbf{Z}_{\text{AF3}} + (\mathbf{1} - \mathbf{M}) \odot \epsilon_{\text{prior}}. \quad (2)$$

Here, $\mathbf{Z}_{\text{AF3}}$ refers to the Apo-MHC (epitope-free state protein) structure predicted by AlphaFold3, and $\mathbf{M} \in \{0, 1\}^{N \times (3+d)}$ is a binary mask (1 for MHC residues, 0 for epitope residues). $\epsilon_{\text{prior}} \in \mathbb{R}^{N \times (3+d)}$ represents the initial noise sampled from a standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Throughout the denoising dynamics ($t = T \rightarrow 0$), the model updates the full state $\mathbf{Z}_t$, allowing the MHC backbone to undergo conformational adjustments to optimize the physicochemical compatibility with the generated epitope.

2.4. Affinity Soft Prompting

We introduce ASP to model binding affinity as a continuous condition within the generative framework. This approach replaces binary classification labels with a scalar representation, allowing the model to distinguish between varying degrees of binding strength.

We map the experimental binding affinity (IC_{50}) to a normalized scalar $s \in [0, 1]$ using a log-linear transformation relative to a null threshold ($\text{IC}_{\text{null}} = 50,000 \text{ nM}$):

$$s = \max\left(0, 1 - \frac{\log_{10}(\text{IC}_{50})}{\log_{10}(\text{IC}_{\text{null}})}\right). \quad (3)$$

To capture high-frequency dependencies across this scalar spectrum, we project s into a high-dimensional feature space using Fourier Positional Embeddings (Tancik et al., 2020):

$$\gamma(s) = \left[\sin(2^k \pi s), \cos(2^k \pi s) \right]_{k=0}^{K-1}. \quad (4)$$

The embedding $\gamma(s)$ is processed by a Multi-Layer Perceptron (MLP) to produce the conditioning vector. This vector is injected into the E(3)-equivariant denoising network, enabling it to condition the noise prediction on the specific affinity level.

2.5. Contrastive Affinity Guidance

While ASP allows the model to condition on high-affinity levels ($s_{\text{high}} \rightarrow 1$), relying solely on the conditional distribution $p_{\theta}(\mathbf{Z}|s_{\text{high}})$ is suboptimal due to the inherent data imbalance in biochemical assays. High-affinity data is typically scarce and expensive to obtain, whereas low-affinity data is abundant. Consequently, the estimated score $\nabla_{\mathbf{Z}} \log p_{\theta}(\mathbf{Z}|s_{\text{high}})$ often suffers from high variance in regions far from the limited positive training samples.

To address this scarcity, we introduce CAG, a sampling strategy designed to improve generation quality by exploiting the information-rich negative manifold. Our core insight is that while the model may struggle to pinpoint exactly what constitutes a good binder due to data sparsity, it can learn robustly what constitutes a non-binder ($s_{\text{low}} \rightarrow 0$) from the abundant negative data. We formulate the guidance not merely as attraction to the positive mode, but as a rectification process using the negative mode. The modified score function is defined as:

$$\nabla_{\mathbf{Z}} \log p_{\gamma}(\mathbf{Z}_t) = \nabla_{\mathbf{Z}} \log p_{\theta}(\mathbf{Z}_t|s_{\text{high}}) + \underbrace{\lambda (\nabla_{\mathbf{Z}} \log p_{\theta}(\mathbf{Z}_t|s_{\text{high}}) - \nabla_{\mathbf{Z}} \log p_{\theta}(\mathbf{Z}_t|s_{\text{low}}))}_{\text{Contrastive Rectification}}. \quad (5)$$

Since $p_{\theta}(\mathbf{Z}|s_{\text{low}})$ is learned from a dense region of the data manifold, its gradient provides a stable, high-confidence directional signal. By subtracting this component, we effectively prune the generative search space, penalizing features associated with non-binding modes. This allows the model to navigate towards high-affinity regions more reliably than by following the sparse positive signal alone, effectively converting the abundance of negative data into a stabilizing force for positive generation.

2.6. Theoretical Motivation

We provide a formal analysis of why the proposed ASP and CAG mechanisms offer superior generative performance compared to standard binary conditioning and positive-only training approaches.

Resolution: Minimizing Conditional Entropy. Standard generative approaches often employ binary labels (binder vs. non-binder), effectively quantizing the continuous affinity space. Let $y \in \{0, 1\}$ be a binary label derived from the continuous affinity s via a threshold τ (i.e., $y = \mathbb{1}(s > \tau)$). The conditional distribution learned by such a model is a marginalization over the high-affinity interval:

$$p_{\theta}(\mathbf{Z} | y = 1) = \int_{\tau}^1 p_{\theta}(\mathbf{Z} | s) p(s | y = 1) ds. \quad (6)$$

Eq. (6) demonstrates that binary conditioning forces the model to approximate a mixture distribution. We quantify

this loss of precision using conditional entropy. Since the binary label y is a deterministic coarsening of s , the data processing inequality implies:

$$H(\mathbf{Z} | s) \leq H(\mathbf{Z} | y), \quad (7)$$

where the inequality strictly holds when the distribution $p(\mathbf{Z}|s)$ is dependent on s . By conditioning on the continuous scalar s via ASP, our model minimizes the conditional entropy of the generated epitopes. This ensures that the generative process targets a specific iso-affinity manifold rather than a diffuse region, theoretically enabling the generation of “best-in-class” rather than “average-good” candidates.

Extrapolation: Geometry of the Log-Odds. The theoretical advantage of CAG lies in its utilization of the negative manifold to approximate the gradient of the binding probability class boundary. Consider the log-odds ratio of an epitope being a high-affinity binder versus a non-binder:

$$\mathcal{L}(\mathbf{Z}) = \log \frac{p_{\theta}(\mathbf{Z} | s_{\text{high}})}{p_{\theta}(\mathbf{Z} | s_{\text{low}})}. \quad (8)$$

The gradient of this ratio, $\nabla_{\mathbf{Z}} \mathcal{L}(\mathbf{Z})$, represents the discriminative direction that maximally separates binders from non-binders.

In positive-only learning, the model estimates the score function $\nabla_{\mathbf{Z}} \log p_{\theta}(\mathbf{Z} | s_{\text{high}})$, which corresponds geometrically to the vector field tangent to the high-affinity manifold. However, without access to negative data, the normal vector pointing towards the manifold from the exterior is undefined. Our CAG term (Eq. 5) explicitly reconstructs this discriminative gradient. By leveraging the abundant low-affinity data to estimate $p_{\theta}(\mathbf{Z} | s_{\text{low}})$, we obtain a high-confidence estimate of the non-binding direction. Subtracting this gradient allows the model to perform an implicit ascent on the binding probability surface:

$$\nabla_{\mathbf{Z}} \log p_{\lambda}(\mathbf{Z}) \approx \nabla_{\mathbf{Z}} \log p_{\theta}(\mathbf{Z} | s_{\text{high}}) + \lambda \nabla_{\mathbf{Z}} \mathcal{L}(\mathbf{Z}). \quad (9)$$

This formulation effectively extrapolates the optimization trajectory beyond the support of the scarce positive training data by utilizing the negative manifold as a repulsive boundary.

3. Experiments

3.1. Experimental Setup

Dataset Construction. We construct our dataset by curating entries from (O’Donnell et al., 2020) and (Marzella et al., 2024). To ensure high data quality, we exclusively select qualitative assays associated with human alleles and filter out invalid data points. The final dataset comprises 100,089 entries, consisting of 44,042 positive binders and

Table 1. Performance comparison between EpiCoCo and baseline methods on the MHC-specific epitope design task. *RLpMIEC results are reported on its supported 13 MHC alleles.

Methods	Uniqueness(%) $\uparrow$	Diversity(%) $\uparrow$	Naturalness(%) $\downarrow$	Instability $\downarrow$	pLDDT ^p $\uparrow$	ipTM $\uparrow$	ΔG $\downarrow$
PepPPO (Bioinfo.23)	<u>99.95</u>	65.38	2.08	37.09	0.8405	0.9132	-29.25
RLpMIEC* (JCIM.24)	99.74	69.51	1.53	41.23	<u>0.8641</u>	<u>0.9215</u>	<u>-37.55</u>
PepMLM (Nat Biotech.25)	100.00	45.64	3.77	42.25	0.6100	0.7435	56.86
DPLM (ICML24)	98.40	70.45	4.26	95.38	0.8636	0.9178	-28.36
PepPrCLIP (Sci.Adv.25)	100.00	<u>71.78</u>	<u>1.27</u>	44.53	0.8529	0.9200	-29.56
PepTune (ICML25)	100.00	61.95	5.17	<u>17.68</u>	0.7598	0.8762	-20.35
MOG-DFM (ICML25)	100.00	67.11	2.55	-56.74	0.5986	0.7576	47.61
PepFlow (ICML24)	81.78	30.36	3.58	36.69	0.8500	0.9208	-23.75
PepGLAD (NIPS24)	99.02	75.41	1.72	35.82	0.7690	0.8744	-13.46
PocketXMol (Cell26)	99.02	68.39	1.36	42.71	0.8428	0.9163	-31.79
EpiCoCo (Ours)	100.00	66.76	1.09	44.22	0.8884	0.9334	-45.20

56,047 negative non-binders. For the training of the generative model, we utilize both the positive and negative samples. Of the 110 distinct MHCs, 100 possess both positive and negative samples, while the remaining 10 solely contain negative samples.

Baselines. To comprehensively evaluate the performance of EpiCoCo, we compare it against competitive baselines across different paradigms. As required by their papers, we provided complete 3D structural inputs for all structure-based models to ensure rigorous reproduction.

Reinforcement Learning (RL): PepPPO (Chen et al., 2023) and RLpMIEC (Deng et al., 2024), the current SOTA methods for specialized MHC-binding peptide generation using reinforcement learning.

Sequence-Conditioned PLMs: PepMLM (Chen et al., 2025a), DPLM (Wang et al., 2024b), and PepPrCLIP (Bhat et al., 2025), which leverage pre-trained protein language models (PLMs) for 1D sequence generation.

Optimization-Based Strategies: PepTune (Tang et al., 2025) and MOG-DFM (Chen et al., 2025b), employing a two-stage approach of unconditional generation followed by classifier guidance.

Structure-Aware paradigm: PepFlow (Li et al., 2024), a flow-matching framework for co-generating structures and sequences; PepGLAD (Kong et al., 2024), a latent diffusion model with VAE for full-atom peptide design under binding sites; and PocketXMol (Peng et al., 2026), a SOTA unified atom-level model for pocket-based small molecule and peptide generation.

Metrics. For a fair comparison among models, we evaluate them on the 100 MHCs that have both positive and negative samples, with each method generating 30 peptide candidates per case, resulting in a total of 3,000 peptides generated. Some key metrics are used for comparison:

(1) **Uniqueness**, the proportion of non-duplicate sequences within the generated pool, reflecting the model’s ability to avoid mode collapse; (2) **Diversity**, the average pairwise dissimilarity between the unique generated sequences, quantifying the breadth of the sequence space explored by the model; (3) **Naturalness**, which measures the distributional similarity to natural epitopes using the mean absolute error (MAE) of amino acid frequencies, with lower values indicating higher biological plausibility. (4) **Instability** (Guruprasad et al., 1990), which measures the stability of the peptide, with lower values indicating higher stability; (5) **pLDDT^p**, which quantifies the confidence in the predicted local structure for each residue of the generated peptide; (6) **ipTM**, which assesses the quality of the predicted interaction interface between the peptide and the MHC, serving as a metric for the structural confidence of the complex formation; (7) **dG_{separated} (ΔG)** (Stranges & Kuhlman, 2013), which estimates the binding free energy (in Rosetta Energy Units (REU)) of the pMHC complex, with lower values indicating stronger physical binding affinity.

More detailed information about the model implementation, baselines, and metrics can be found in the Appendix Sections B and C.

3.2. Experimental Results

Table 1 summarizes the quantitative performance comparison where EpiCoCo establishes a new SOTA by achieving a binding free energy (ΔG) of -45.20 REU on average. This represents a significant 23% improvement over the strongest baseline, RLpMIEC, which reaches -37.55 REU but is restricted to only 13 common alleles. In contrast, EpiCoCo demonstrates robust generalization across the full dataset of 100 MHCs, validating its capability to model diverse targets without allele-specific fine-tuning. Complementing this energetic performance, the generated candidates exhibit

high structural plausibility by attaining the highest confidence scores among all methods. Specifically, EpiCoCo achieves an ipTM of 0.9334, which indicates superior predicted quality of the pMHC interaction interface, alongside a residue-level confidence (pLDDT^p) of 0.8884.

The comparison with PepFlow offers valuable insights into the impact of conditioning strategies. PepFlow, which conditions generation on a fixed MHC backbone, records a uniqueness score of 81.78% and a diversity score of 30.36%. Its binding energy of -23.75 REU suggests that relying on a static scaffold may restrict exploration of the solution space, potentially limiting the discovery of binders that depend on structural adaptation. In comparison, by operating on the joint manifold where the MHC is treated as a dynamic context, EpiCoCo achieves 100.00% uniqueness and a diversity of 66.76%, comparable to sequence-only models. These results indicate that our framework effectively leverages geometric constraints to optimize affinity while maintaining generative variety.

In terms of sequence properties, EpiCoCo achieves the best naturalness MAE of 1.09. This low score indicates that the generated sequences statistically resemble the distribution of natural epitopes and thus possess high biological validity. While PepTune and MOG-DFM achieve lower instability scores, their positive binding energy implies the generation of peptides that are stable in isolation but may fail to bind the receptors. This discrepancy arises because high-affinity binders often require hydrophobic anchor residues to bury into the MHC pocket, which inherently decreases the intrinsic stability of peptides (Wieczorek et al., 2017b; Madden, 1995). EpiCoCo successfully navigates this trade-off by minimizing the naturalness MAE while optimizing binding energy, thereby capturing the intrinsic physicochemical principles of antigen presentation.

3.3. Ablation Study

To dissect the contribution of each component in EpiCoCo, we evaluate the binding energy ΔG across distinct model configurations. We generate 30 epitopes for each of the 100 MHC alleles that possess both positive and negative samples, resulting in a total of 3,000 generated candidates. Table 2 illustrates the impact of MHC-Context Co-Modeling, ASP, and CAG. Since CAG is an inference-time guidance mechanism dependent on the prompting module ASP, a configuration featuring CAG without ASP is technically infeasible and is excluded. More results are provided in Appendix Section D.

Co-modeling yields substantial gain in binding affinity.

Across all settings, the co-modeling strategy consistently outperforms static MHC conditioning. Rigid conditions impose strict constraints, narrowing the search space. By

Table 2. Ablation study on the contribution of core components in terms of binding energy (ΔG). ASP: Affinity Soft Prompting; CAG: Contrastive Affinity Guidance.

Co-Modeling	ASP	CAG	$\Delta G \downarrow$
			-32.75
	✓		-41.62
	✓	✓	-43.25
✓			-41.38
✓	✓		-42.67
✓	✓	✓	-45.20 (Ours)

jointly updating the epitope and MHC, EpiCoCo effectively relaxes the structure to optimize binding modes. This advantage is most striking in the unguided baseline, where co-modeling outperforms static conditioning by nearly **9 REU** ($\sim 26\%$). While the guidance from ASP and CAG can partially compensate for a rigid scaffold by forcing sequence adaptation, the massive performance gap in the unguided setting underscores that conformational flexibility is a fundamental prerequisite for naturally modeling high-affinity interactions.

ASP consistently reduces binding energy via continuous conditioning.

The comparison between the unguided baseline and the ASP-enabled model demonstrates the efficacy of affinity soft prompting. ASP yields energy improvements across both settings. By injecting the continuous affinity signal s into the diffusion backbone, ASP enables the network to internalize fine-grained affinity-structure correlations during training, capturing the subtle physicochemical signatures.

CAG further enhances affinity by suppressing non-binding patterns.

The comparison between the ASP-enabled models and the ASP + CAG models shows that CAG contributes in binding energy in both settings, improving our full model by **2.53 REU** ($\sim 6\%$). While standard sampling tends to stay within the training distribution, CAG leverages repulsive gradients from non-binders to guide trajectories away from moderate-affinity regions. This contrastive mechanism allows the model to extrapolate beyond training modes and discover epitopes with higher affinity levels than those observed during training.

4. Case Study

4.1. Analysis of Sequence Motif Profiles and Structural Co-Adaptation on HLA-A*02:01

Building on the quantitative superiority observed in sequence-related metrics (including naturalness, uniqueness, and diversity), we now conduct a qualitative analysis to

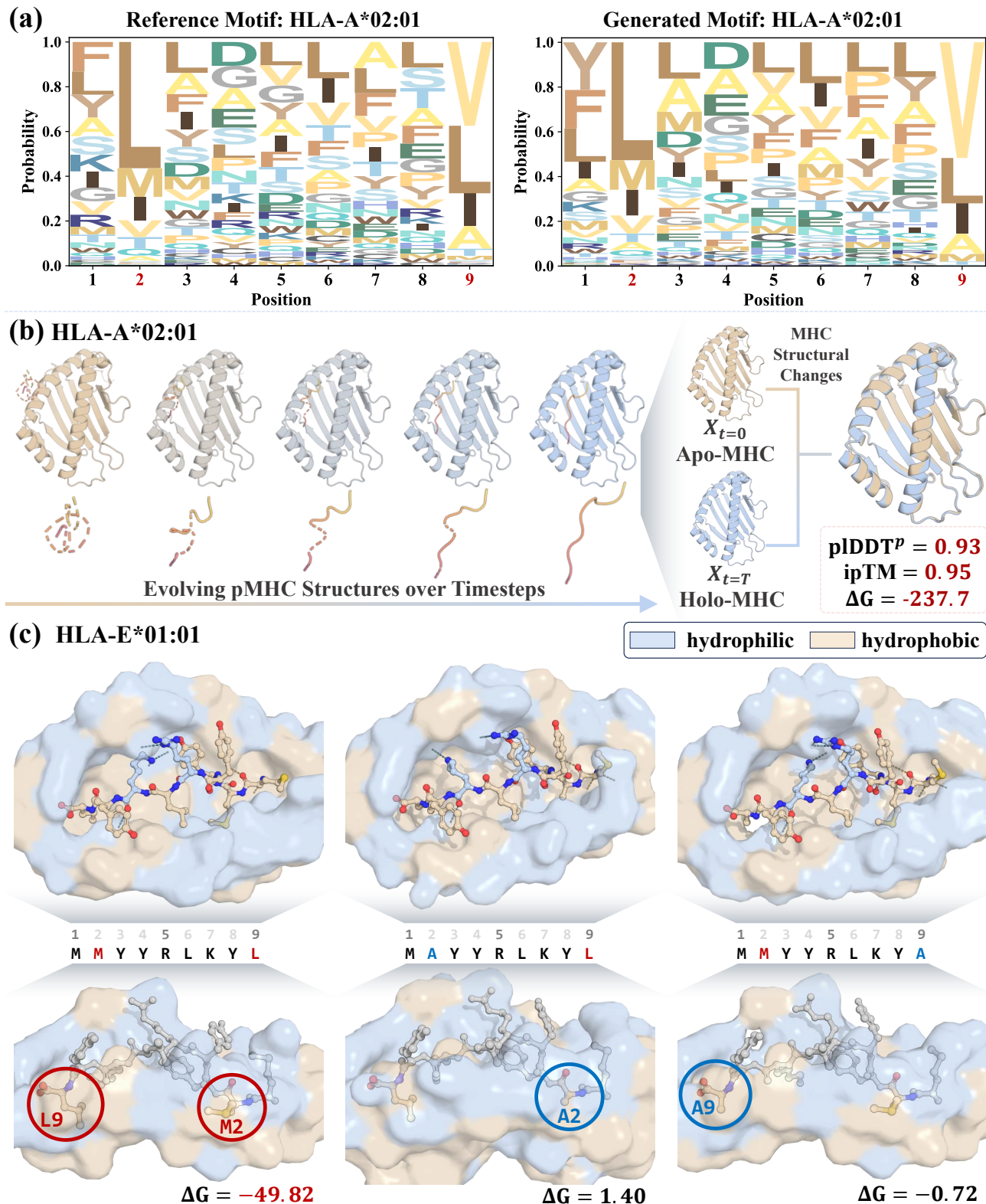


Figure 3. Visual analysis of pMHC co-modeling and generalization. (a) Sequence motif comparison between reference and generated epitopes (HLA-A*02:01). (b) **Dynamic Adaptation.** The generative trajectory (Left) illustrates the joint evolution of the epitope and MHC conformation. The overlay (Right) of initial and final states highlights α -helix adjustments that accommodate the epitope to optimize binding affinity. (c) **One-Negative-Shot Generation (HLA-E*01:01).** Top: Physicochemical complementarity showing precise alignment of hydrophobic and hydrophilic residues within MHC pockets. Bottom: *In silico* mutational analysis confirms key anchor motifs; P2 (M→A) or P9 (L→A) mutations disrupt deep geometric locking, causing drastic degradation in binding energy.

illustrate the biological validity of the generated sequences. Focusing on the prevalent allele HLA-A*02:01, we examine whether EpiCoCo successfully captures underlying structural determinants rather than merely memorizing statistical patterns.

Sequence Motif Fidelity. As shown in Figure 3a, the sequence logos generated by EpiCoCo exhibit high consistency with the ground truth preferences of HLA-A*02:01, particularly the conservation of anchors at P2 (Leucine or Methionine) and P9 (Valine or Leucine). This fidelity highlights the efficacy of our geometric modeling. By integrating spatial priors such as inter-atomic distances and angles into the generative backbone, EpiCoCo effectively captures the intricate dependencies between residue positions. This allows the model to internalize the physicochemical constraints of the binding pocket, ensuring that generated sequences adhere to the specific anchor motifs requisite for stable binding.

Structural Plausibility. Beyond sequence validity, we observe a phenomenon of *structural co-adaptation* during the generation process. As shown in Figure 3b, the generative trajectory reveals synchronized micro-adjustments within the peptide-binding groove. This mimics the biological mechanism of induced-fit, where the receptor spatially reorients to accommodate the ligand. The physical plausibility of this dynamic interface is corroborated by high confidence scores (pLDDT^P: 0.93, ipTM: 0.95) and a favorable binding energy ($\Delta G = -237.7$ REU), confirming that the model optimizes the joint energy landscape of the complex.

4.2. Investigation of Generalization Capabilities on Few-Negative-Shot Alleles

To assess the generalization capability under extreme data scarcity, we analyze the 10 MHC alleles absent from the positive training set and limited to sparse negative samples (the *Few-Negative-Shot* scenario).

Quantitative Analysis and Negative-Informed Mechanism. As detailed in Table 3, we compare EpiCoCo against PepPPO and PepPrCLIP, which are identified as the top-performing baselines in our primary evaluation (Table 1). EpiCoCo achieves significantly stronger binding affinities compared to these competitors across all tested targets. This superior performance validates the effectiveness of our negative-informed generation strategy enabled by ASP and CAG. Conventional generative paradigms predominantly rely on modeling the distribution of positive binders and thus struggle to define a valid optimization direction when positive examples are entirely absent. In contrast, our framework leverages the available sparse negative data to map the low-affinity landscape. During inference, CAG uti-

Table 3. Binding free energy comparison across 10 few-negative-shot alleles (30 candidates each).

MHC	ΔG_{PepPPO}	$\Delta G_{\text{PepPrCLIP}}$	$\Delta G_{\text{EpiCoCo}}$
HLA-A*02:04	<u>-35.16</u>	9.35	-39.36
HLA-B*15:10	<u>-27.79</u>	<u>-28.06</u>	-70.92
HLA-B*27:02	<u>-25.71</u>	-25.61	-45.85
HLA-B*27:03	<u>-26.53</u>	-23.49	-41.20
HLA-B*27:04	-24.84	<u>-28.89</u>	-40.36
HLA-B*27:09	-26.24	<u>-30.75</u>	-37.82
HLA-B*27:10	-28.64	<u>-29.54</u>	-38.00
HLA-B*39:09	<u>-36.98</u>	-32.36	-48.14
HLA-C*08:03	<u>-23.57</u>	-32.32	-42.12
HLA-E*01:01	<u>-29.49</u>	-28.52	-48.48

lizes this information to apply a repulsive gradient, actively guiding the generative trajectory away from non-binding modes. This mechanism allows the model to extrapolate towards the high-affinity manifold by effectively inverting the discriminative signals from negative samples.

MD Simulations. To strengthen claims, we conduct comprehensive molecular dynamics (MD) simulations on the few-negative-shot generated complexes. Specifically, for each target, we first select the top 3 generated peptides based on Rosetta reference energy (ΔG), and then perform a 50 ns MD simulation. As shown in Table 4, generated peptides remain firmly anchored in the binding groove with Ligand In-Pocket RMSD consistently below 1.35 Å, reaching sub-angstrom stability (< 0.50 Å) for alleles like HLA-B*27:04. Receptor backbones similarly maintain structural integrity (RMSD 1.4–3.5 Å). Furthermore, the complexes exhibit robust intermolecular hydrogen-bond networks: while HLA-A and HLA-E variants baseline at 6–7 mean H-bonds, HLA-B and HLA-C form notably denser networks averaging 10–15 H-bonds (peaking at 21 for HLA-B*27:02). Finally, MM/GBSA calculations confirm highly favorable thermodynamic stability, with deep binding energies ranging from -67.30 kcal/mol (HLA-A*02:02) to -142.37 kcal/mol (HLA-B*39:09). This confirms that EpiCoCo generates epitopes with true thermodynamic binding capacity rather than merely superficial geometric fits.

Anchor Residue Verification via *In Silico* Mutagenesis. We further examine the generalization capability on HLA-E*01:01 to assess biological validity under data-scarce conditions. This non-classical MHC-Ib allele serves as a critical ligand for inhibitory receptors on NK cells and represents a conserved target for universal cancer vaccines (Schoufour et al., 2025; Früh et al., 2026), while the dataset contains only a single negative sample for this target. Despite this scarcity, EpiCoCo generates high-affinity epitopes with an average binding energy of -48.48 REU.

Table 4. MD simulation results for few-negative-shot generated peptide-MHC complexes (50 ns). For each allele, the top-3 generated complexes out of 30 (by Rosetta ΔG) are selected.

Complex	Rec_RMSD (Å)	Lig_InPocket_RMSD (Å)	Inter_HBond (Mean/Max)	GBSA ΔG (kcal/mol)	PBSA ΔG (kcal/mol)
HLA-A*02:04	1.70 $\pm$ 0.12	0.73 $\pm$ 0.16	6.24/12	-68.39 $\pm$ 11.95	-45.07 $\pm$ 13.57
HLA-B*15:10	2.64 $\pm$ 0.72	0.61 $\pm$ 0.06	11.76/18	-102.14 $\pm$ 9.17	-87.10 $\pm$ 10.81
HLA-B*27:02	2.89 $\pm$ 0.18	1.32 $\pm$ 0.27	15.36/21	-131.95 $\pm$ 7.66	-117.08 $\pm$ 10.07
HLA-B*27:03	2.17 $\pm$ 0.36	0.64 $\pm$ 0.13	11.23/18	-129.07 $\pm$ 12.09	-109.42 $\pm$ 9.98
HLA-B*27:04	1.94 $\pm$ 0.19	0.48 $\pm$ 0.07	12.05/20	-112.42 $\pm$ 21.24	-93.17 $\pm$ 28.61
HLA-B*27:09	2.53 $\pm$ 0.32	0.75 $\pm$ 0.14	11.50/17	-96.02 $\pm$ 7.36	-88.27 $\pm$ 8.36
HLA-B*27:10	3.43 $\pm$ 0.17	0.71 $\pm$ 0.11	12.09/20	-101.33 $\pm$ 12.64	-87.18 $\pm$ 12.50
HLA-B*39:09	2.81 $\pm$ 0.44	0.55 $\pm$ 0.13	10.82/16	-142.37 $\pm$ 7.63	-115.94 $\pm$ 7.41
HLA-C*08:03	1.47 $\pm$ 0.21	0.49 $\pm$ 0.18	11.77/18	-109.30 $\pm$ 9.31	-93.00 $\pm$ 11.29
HLA-E*01:01	1.99 $\pm$ 0.22	1.25 $\pm$ 0.18	7.18/12	-92.93 $\pm$ 8.19	-71.05 $\pm$ 9.69
HLA-A*02:02	1.72 $\pm$ 0.19	1.06 $\pm$ 0.19	7.02/14	-67.30 $\pm$ 10.06	-41.16 $\pm$ 10.39

As shown in Figure 3c, the generated sequence strictly adheres to the HLA-E binding motif by featuring Methionine at P2 and Leucine at P9 (Ruibal et al., 2020). We validate the energetic contribution of these anchors via *in silico* mutational scanning, substituting them with all 19 other amino acids; the results reveal a consistent energy penalty across all variations (see Appendix Figure 4), confirming the optimality of our predictions. Structural visualization (Figure 3c, middle and right) further demonstrates that Alanine mutants fail to establish optimal contacts within the specificity pockets, resulting in a drastic loss of affinity ($\Delta\Delta G$ of P2_{M→A}: +51.22 REU; P9_{L→A}: +49.10 REU). This high affinity is driven by exact physicochemical complementarity, where hydrophobic epitope residues are deeply buried within non-polar MHC pockets. This showcases the model’s ability to simulate structural adaptation, allowing the HLA-E binding groove to undergo geometric adjustments to accommodate the steric requirements of the anchor side chains.

5. Conclusion

In this work, we presented EpiCoCo, a framework that redefines *de novo* epitope design by integrating contrastive affinity guidance into pMHC complex co-modeling. By leveraging the disparity between high- and low-affinity distributions, our approach effectively navigates the complex energetic landscape of biochemical interactions, capturing the intricate induced-fit dynamics between MHC receptors and potential binders. This methodology ensures robust generation quality, yielding candidates with SOTA binding affinity and structural plausibility. While the translation of these computational findings invites future large-scale wet-lab validation to fully map the immunogenic landscape, the framework demonstrates exceptional structural consistency and binding plausibility. Ultimately, EpiCoCo highlights the potential of geometric deep learning for controllable

molecular generation, offering a promising computational tool to facilitate peptide-based therapeutic design.

Impact Statement

EpiCoCo enables the efficient computational design of high-affinity epitopes for personalized immunotherapy, accelerating vaccine development and cancer treatment while reducing experimental costs and animal testing.

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A. Related Works

The transition from binding classification to generative design represents a key shift in computational immunology. We categorize recent advances by their representation of the MHC context and their learning objectives.

Sequence-Centric Modeling. The majority of current generative models focus on sequence-centric approaches, treating epitope design as a conditional language task. Methods such as PepMLM (Chen et al., 2025a) and PepPrCLIP (Bhat et al., 2025) fine-tune protein language models to reconstruct epitopes based on the MHC sequence. While this captures co-evolutionary patterns, it ignores the physical reality of binding. By treating the MHC as a textual context, these models overlook geometric boundaries. Consequently, sequence-only priors often yield candidates that are linguistically plausible but fail to fit physically or position anchor residues correctly within the MHC groove.

Static MHC Geometric Conditioning. To resolve the challenge of geometric complementarity, the field has recently begun to pivot toward structure-conditioned generation. Two-stage workflows, such as PMGen (Asgary et al., 2025), generate sequences first and validate them via external folders like AlphaFold. However, this separates the design process from physical validation. More advanced geometric models, including PepFlow (Li et al., 2024), PepGLAD (Kong et al., 2024), ApexGen (Xia & de la Fuente-Nunez, 2025), and Pocket2Mol (Feng et al., 2024) directly integrate structural inputs. Similarly, strategies like PepMimic (Kong et al., 2025) focus on replicating binding interfaces within these static boundaries. These models typically encode the MHC as a fixed/rigid geometric condition in $\mathbb{R}^3$. This static conditioning simplifies binding physics by assuming a frozen MHC backbone. Biological evidence shows that high-affinity binding is an induced-fit process, involving conformational adjustments of the MHC (Wieczorek et al., 2017a). Unified modeling that captures these joint fluctuations is essential, yet it remains a rarely explored field.

Property-Guided Controllable Generation. Current approaches for optimizing molecular properties largely fall into two categories: Reinforcement Learning (RL) and controllable generative models. While RL has shown success in sequence design (Chen et al., 2023; Deng et al., 2024), it often suffers from sparse rewards and training instability, especially in high-dimensional structure spaces.

In contrast, controllable generative models, particularly those leveraging classifier guidance (Tang et al., 2025; Chen et al., 2025b), offer more stable gradients for optimization. Recent general molecular diffusion models have advanced this direction. KGDiff (Qian et al., 2023) employed continuous property guidance in the field of molecule design.

Despite these advances in general drug design, pMHC binding modeling has yet to fully adopt these continuous, full-spectrum strategies. Existing structure-based MHC methods primarily rely on binary class conditioning (binder/non-binder), missing the precise guidance offered by continuous affinity landscapes and the discriminative power of negative samples.

B. Details of Model

In this section, we detail the specific implementation of the EpiCoCo framework. We describe the data processing pipeline, the specific architectural modifications to the equivariant backbone, and the exact training objectives used to model the pMHC complex.

B.1. Data Representation

The system state at diffusion step t is represented as a geometric graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with state tuple $\mathbf{Z}_t = (\mathbf{X}_t, \mathbf{H}_t, \mathbf{E}_t)$. Here, $\mathbf{X}_t \in \mathbb{R}^{N \times 3}$ represents the continuous Cartesian coordinates of the residues. $\mathbf{H}_t \in \mathbb{R}^{N \times d}$ denotes the residue features, which are treated as continuous vectors in a Euclidean space rather than discrete categorical indices. $\mathbf{E}_t$ represents the edge attributes connecting interacting residues. The set $\mathcal{V}$ encompasses the entire pMHC complex, partitioned into MHC residues and epitope residues.

B.2. Network Architecture and Input Featurization

B.2.1. CONTEXT-AWARE NODE INITIALIZATION

To condition the generation on biological constraints, we initialize node features by fusing structural identity with rich contextual embeddings. For a given node i , the input feature $\mathbf{h}_i^{\text{in}}$ is constructed as:

$$\mathbf{h}_i^{\text{in}} = [(\mathbf{W}_{\text{base}} \mathbf{z}_i + \mathbf{E}_{\text{type}}(c_i)) \parallel \mathbf{h}_i^{\text{ctx}}], \quad (10)$$

where $\mathbf{z}_i$ is the continuous amino acid vector, $c_i \in \{\text{MHC}, \text{Epitope}\}$ is the chain indicator, $\mathbf{W}_{\text{base}}$ is a learnable linear projection, and $\parallel$ denotes concatenation. The context vector $\mathbf{h}_i^{\text{ctx}}$ integrates four information sources:

$$\mathbf{h}_i^{\text{ctx}} = \Phi_{\text{ctx}}(\text{TimeEmb}(t) \oplus \mathbf{f}_{\text{ESM},i} \oplus \bar{\mathbf{f}}_{\text{MHC}} \oplus \mathbf{c}_{\text{aff}}). \quad (11)$$

Here, $\text{TimeEmb}(t)$ is the sinusoidal embedding of the timestep, $\mathbf{f}_{\text{ESM},i}$ is the residue-level ESM-2 embedding, $\bar{\mathbf{f}}_{\text{MHC}}$ is the pooled global ESM-2 embedding of the MHC allele (broadcast to all nodes), and $\mathbf{c}_{\text{aff}}$ is the affinity soft prompt. Φ_{ctx} is a Multi-Layer Perceptron (MLP) that projects these concatenated features to the hidden dimension.

B.2.2. HIGHWAY GATING MECHANISM

To mitigate the vanishing gradient problem inherent in deep graph networks and to facilitate the flow of information through the 12-layer backbone, we implement a specific **Highway Gating** mechanism within each node update block. Unlike standard residual connections, this allows the network to learn an adaptive blend between the input and the update. The update rule for node features at layer l is defined as:

$$\begin{aligned} \mathbf{g}_i &= \sigma(\mathbf{W}_{\text{gate}}\mathbf{h}_i^l + \mathbf{b}_{\text{gate}}), \\ \mathbf{h}_i^{l+1} &= \mathbf{h}_i^l + \mathbf{g}_i \odot (\Phi_{\text{node}}(\mathbf{h}_i^l, \mathbf{m}_i) - \mathbf{h}_i^l). \end{aligned} \quad (12)$$

Here, σ is the sigmoid activation function, $\odot$ denotes element-wise multiplication, and $\mathbf{m}_i$ is the aggregated message from neighbors. Φ_{node} represents the node update MLP. A crucial implementation detail is the initialization of the bias vector $\mathbf{b}_{\text{gate}}$. We initialize $\mathbf{b}_{\text{gate}}$ to -3.0 , which biases the gate $\mathbf{g}_i$ toward zero at the start of training. This effectively forces the layer to approximate an identity mapping initially, stabilizing the optimization of the deep architecture.

B.3. Affinity Soft Prompting (ASP)

We implement ASP to inject fine-grained affinity control. We first normalize the experimental IC_{50} value to a scalar $s \in [0, 1]$:

$$s = \text{clamp}\left(1 - \frac{\log_{10}(\text{IC}_{50})}{\log_{10}(50,000)}, 0, 1\right). \quad (13)$$

To enable the network to resolve high-frequency changes in affinity, we project s using Fourier feature mapping before passing it through a learnable MLP:

$$\begin{aligned} \gamma(s) &= \left[\sin(2^k \pi s), \cos(2^k \pi s) \right]_{k=0}^{K-1}, \\ \mathbf{c}_{\text{aff}} &= \Phi_{\text{aff}}(\gamma(s)), \end{aligned} \quad (14)$$

where K is the number of frequency bands. The resulting embedding $\mathbf{c}_{\text{aff}}$ is added to the node context.

B.4. Training Objectives

We optimize the model using a weighted sum of Mean Squared Error (MSE) losses across three modalities: structure, sequence, and topology. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{pos}}\mathcal{L}_{\text{pos}} + \lambda_{\text{seq}}\mathcal{L}_{\text{seq}} + \lambda_{\text{edge}}\mathcal{L}_{\text{edge}}. \quad (15)$$

Structural Loss ($\mathcal{L}_{\text{pos}}$). We supervise the coordinate predictions by comparing the predicted noise on positions against the ground truth Gaussian noise:

$$\mathcal{L}_{\text{pos}} = \mathbb{E} [\|\epsilon^{\text{pos}} - \epsilon_{\theta}^{\text{pos}}(\mathbf{Z}_t, t, \mathbf{c})\|^2], \quad (16)$$

where ϵ^{pos} is the sampled Gaussian noise added to the coordinates.

Sequence Loss ($\mathcal{L}_{\text{seq}}$). Since we treat amino acids as continuous vectors, we apply MSE loss on the residue features:

$$\mathcal{L}_{\text{seq}} = \mathbb{E} [\|\epsilon^{\text{seq}} - \epsilon_{\theta}^{\text{seq}}(\mathbf{Z}_t, t, \mathbf{c})\|^2], \quad (17)$$

where ϵ^{seq} is the noise added to the continuous amino acid embeddings.

Edge Type Loss ($\mathcal{L}_{\text{edge}}$). To explicitly learn the topology of the epitope, we predict the edge types between residues. We treat edge types as continuous one-hot vectors and apply MSE loss:

$$\mathcal{L}_{\text{edge}} = \mathbb{E} \left[\|\epsilon^{\text{edge}} - \epsilon_{\theta}^{\text{edge}}(\mathbf{Z}_t, t, \mathbf{c})\|^2 \right]. \quad (18)$$

Here, the ground truth is a one-hot encoding where the active index corresponds to edges within the epitope chain (i.e., $j = i \pm 1$ within the epitope), while other indices represent non-bonded interactions or MHC edges.

B.5. Pseudocode of EpiCoCo Algorithm

Algorithm 1 summarizes the training procedure utilizing Affinity Soft Prompting (ASP) and the inference procedure employing Contrastive Affinity Guidance (CAG).

Algorithm 1 EpiCoCo: Affinity-Conditioned Training and Guided Inference

```

1: # Phase 1: Training (Global Joint Diffusion)
2: while not converged do
3:   Sample data  $(\mathbf{Z}_0, \text{IC}_{50})$  from dataset  $\mathcal{D}$     $\{\mathbf{Z}_0 = [\mathbf{X}_{\text{pos}}, \mathbf{H}_{\text{seq}}, \mathbf{E}_{\text{edge}}]\}$ 
4:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
5:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
6:    $\mathbf{Z}_t \leftarrow \sqrt{\bar{\alpha}_t} \mathbf{Z}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$    {Global noise injection}
7:   # Affinity Soft Prompting (ASP)
8:    $s \leftarrow \text{Clamp}(1 - \log_{10}(\text{IC}_{50}) / \log_{10}(\text{IC}_{\text{null}}))$ 
9:    $\mathbf{c}_{\text{aff}} \leftarrow \Phi_{\text{aff}}(\text{Fourier}(s))$ 
10:   $\mathbf{h}^{\text{ctx}} \leftarrow \Phi_{\text{ctx}}(\text{Time}(t) \oplus \mathbf{F}_{\text{ESM}} \oplus \mathbf{c}_{\text{aff}})$ 
11:  # Predict noise and optimize
12:   $\hat{\epsilon} \leftarrow \epsilon_{\theta}(\mathbf{Z}_t, t, \mathbf{c}_{\text{ctx}})$ 
13:   $\mathcal{L} \leftarrow \|\epsilon - \hat{\epsilon}\|^2$    {MSE on Pos, Seq, and Edge types}
14:  Update  $\theta$  via gradient descent  $\nabla_{\theta} \mathcal{L}$ 
15: end while

16: # Phase 2: Inference (Scaffold-Initialized Generation)
17: Input: Apo-MHC structure  $\mathbf{Z}_{\text{MHC}}$ , Mask  $\mathbf{M}$  (1 for MHC, 0 for Epitope)
18: # Scaffold Initialization
19:  $\epsilon_{\text{prior}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
20:  $\mathbf{Z}_T \leftarrow \mathbf{M} \odot \mathbf{Z}_{\text{MHC}} + (\mathbf{1} - \mathbf{M}) \odot \epsilon_{\text{prior}}$ 
21: # Pre-compute Affinity Prompts for Guidance
22:  $\mathbf{c}_{\text{aff}}^{\text{high}} \leftarrow \Phi_{\text{aff}}(\gamma(s_{\text{high}}))$ 
23:  $\mathbf{c}_{\text{aff}}^{\text{low}} \leftarrow \Phi_{\text{aff}}(\gamma(s_{\text{low}}))$ 
24: for  $t = T, \dots, 1$  do
25:   # Contrastive Affinity Guidance (CAG)
26:    $\hat{\epsilon}_{\text{pos}} \leftarrow \epsilon_{\theta}(\mathbf{Z}_t, t, \mathbf{H}_{\text{high}})$ 
27:    $\hat{\epsilon}_{\text{neg}} \leftarrow \epsilon_{\theta}(\mathbf{Z}_t, t, \mathbf{H}_{\text{low}})$ 
28:    $\hat{\epsilon}_{\text{guided}} \leftarrow \hat{\epsilon}_{\text{pos}} + \lambda(\hat{\epsilon}_{\text{pos}} - \hat{\epsilon}_{\text{neg}})$ 
29:   # Reverse Diffusion Step (Update Sequence & Structure)
30:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$  else  $\mathbf{0}$ 
31:    $\mathbf{Z}_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{Z}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \hat{\epsilon}_{\text{guided}} \right) + \sigma_t \mathbf{z}$ 
32: end for
33: Return Generated Epitope Structure & Sequence from  $\mathbf{Z}_0$ 

```

C. Details of Experimental Setup

C.1. Metrics

To provide a holistic assessment of the generated epitopes, we employ a comprehensive evaluation protocol encompassing sequence, structure, and function domains.

Sequence-Level Evaluation. We assess the generative diversity and biological validity using three metrics:

- **Diversity** quantifies the variation within the generated set for a specific MHC allele. It is computed based on pairwise alignment scores using the `PairwiseAligner` from Biopython (Cock et al., 2009). A higher value implies that the model explores a broader chemical space rather than converging to similar patterns.
- **Uniqueness** indicates the redundancy of the generation process. It is defined as the fraction of distinct non-duplicated sequences relative to the total number of generated candidates.
- **Naturalness** measures the distributional distance between the amino acid composition of generated peptides and real-world epitopes. We calculate this as the mean absolute error (MAE) between the residue frequencies of the generated batch and the sample set of natural binders. Lower scores denote a higher statistical resemblance to the natural protein grammar.

Structure-Level Evaluation. We utilize `tFold-TCR` (Wu et al., 2025) to predict the 3D conformation of the pMHC complexes. Based on these predictions, we evaluate:

- **pLDDT (Predicted Local Distance Difference Test)** serves as a proxy for the local structural reliability. It represents the model’s per-residue confidence in the predicted coordinates.
- **pLDDT^p** restricts the pLDDT metric to the peptide chain, focusing specifically on the confidence of the generated epitope geometry.
- **ipTM (Interface Predicted Template Modeling)** assesses the quality of the intermolecular interaction. This metric quantifies the accuracy of the predicted interface between the peptide and the MHC receptor relative to learned structural templates.

Function-Level Evaluation. We focus on stability and binding potency to estimate therapeutic potential:

- **Instability** (Guruprasad et al., 1990) evaluates the chemical stability of the peptide sequence using `ProteinAnalysis` from Biopython (Cock et al., 2009). For peptide therapeutics, lower scores are desirable as they correlate with resistance to degradation and better intrinsic stability prior to receptor binding.
- **dG_{separated} (ΔG)** estimates the binding free energy (measured in Rosetta Energy Units, REU) of the complex. To ensure robust estimation, we implement a rigorous multi-stage protocol: (1) Initial conformations are generated via `tFold-TCR` (Wu et al., 2025); (2) Structures undergo fine-grained refinement using `FlexPepDock` (Raveh et al., 2010), producing 10 decoys per complex (5 high-resolution and 5 low-resolution); (3) Binding energetics are computed using the `InterfaceAnalyzer` (Stranges & Kuhlman, 2013) in Rosetta; (4) The optimal (minimum) energy among the decoys is reported. Lower ΔG values indicate stronger physical affinity.

In summary, sequence metrics ensure the model adheres to biological constraints, while structure and function metrics validate the geometric plausibility and binding potency, collectively identifying candidates with high promise for forming stable pMHC complexes.

C.2. Baselines

To comprehensively evaluate the effectiveness of EpiCoCo, we compare it against a diverse set of representative baselines. We categorize these methods into two primary groups: sequence-centric approaches (which generate 1D amino acid sequences) and structure-aware approaches (which explicitly model geometric coordinates).

Sequence-Centric Paradigms. The following three categories operate strictly within the 1D sequence modality, treating the MHC context either as a text string or a latent vector without modeling explicit 3D interactions:

- **Reinforcement Learning.** We include **PepPPO** (Chen et al., 2023) and **RLpMIEC** (Deng et al., 2024), which formulate epitope generation as a sequential token decision process. **PepPPO** supports conditioning on arbitrary MHC sequences, allowing evaluation across our full dataset. In contrast, **RLpMIEC** relies on allele-specific classifiers and is restricted to 13 common variants. Thus, we evaluate it exclusively on its supported alleles.

- **PLM-Based Generation.** We evaluate **DPLM** (Wang et al., 2024b), **PepMLM** (Chen et al., 2025a), and **PepPrCLIP** (Bhat et al., 2025). These methods leverage pre-trained protein language models (PLMs) to generate epitope sequences conditioned solely on the **linear MHC sequence**. We fine-tune these models on our curated dataset to adapt their extensive prior knowledge to the specific target-conditioned generation task.
- **Two-Stage Optimization.** **PepTune** (Tang et al., 2025) and **MOG-DFM** (Chen et al., 2025b) represent a generate-then-optimize paradigm in the sequence space. They first perform unconditional sequence generation to capture the basic distribution, followed by guidance-based optimization to satisfy functional constraints. For fair benchmarking, we align their guidance objectives using the same affinity and stability predictors employed in our framework.

Structure-Aware Paradigm. Distinct from the sequence-only methods above, we compare against **PepFlow** (Li et al., 2024), a SOTA multimodal flow-matching framework. PepFlow goes beyond 1D tokens by explicitly generating full-atom coordinates for the peptide on the E(3) manifold. However, unlike our co-design approach, it conditions the generation on a **fixed MHC backbone structure**, serving as a critical baseline to assess the necessity of modeling dynamic receptor flexibility.

C.3. Training Details and Hyperparameters

In this section, we provide a comprehensive overview of the computational infrastructure and optimization strategies used to train EpiCoCo.

Implementation and Infrastructure The framework was implemented using PyTorch and PyTorch Geometric. All experiments were conducted on a workstation equipped with a **single NVIDIA GeForce RTX 3090 (24GB) GPU**. Notably, our implementation is highly efficient: training for the full 500,000 iterations requires approximately **7 hours**. To maximize memory efficiency and throughput, we utilized Automatic Mixed Precision (AMP) with `bfloat16` data types.

Dataset and Utilization. The final curated dataset comprises 100,089 pMHC entries, consisting of 44,042 positive binders and 56,047 negative non-binders. Across the 110 distinct MHC alleles, 100 possess both positive and negative samples, while the remaining 10 contain solely negative samples. We utilize the complete dataset for training. As EpiCoCo is a generative framework, utilizing the full dataset does not constitute data leakage. Instead, the validity of the model is assessed by its ability to synthesize new, biologically valid compounds.

Optimization Protocol. We trained the model using the AdamW optimizer with a base learning rate of 1×10^{-4} and weight decay of 1×10^{-6} . We employed a `ReduceLROnPlateau` scheduler to dynamically adjust the learning rate based on loss convergence. The model was trained for a total of 500,000 iterations. For inference and evaluation, we utilized the final checkpoint to ensure the denoising network had fully converged to the target noise schedule.

Hyperparameter Configurations Detailed configurations for the model architecture, diffusion process, and optimization parameters are summarized in Table 5.

D. Details of Results

D.1. Energetic Optimality of Anchor Residues

To rigorously verify that our model has learned the underlying biophysical constraints of the HLA-E complex rather than merely memorizing sequence statistics, we conducted a systematic perturbation analysis. We focused on the critical anchor positions P2 and P9, which are the primary determinants of peptide-MHC stability. Starting with the generated high-affinity binder **MMYYRLKYL**, we computationally mutated these positions to every other naturally occurring amino acid and relaxed the resulting structures to measure the change in Binding Free Energy (ΔG).

Figure 4 visualizes this energetic landscape. The results provide strong evidence for the structural validity of our generation:

P2 Specificity (Methionine): The left panel demonstrates that the generated Methionine (M) is the unique global minimum ($\Delta G \approx -49.8$ REU). The energy gap to the next best hydrophobic alternatives (e.g., Leucine or Phenylalanine) is substantial (> 15 REU), indicating a highly specific stereochemical fit. Notably, substituting with hydrophilic residues (teal bars) or the small neutral Alanine results in a catastrophic loss of affinity (shifting ΔG near or above zero), confirming that the P2 pocket strictly requires a bulky hydrophobic side chain to exclude water and maximize van der Waals contact.

P9 Specificity (Leucine): Similarly, the right panel confirms Leucine (L) as the optimal anchor for the C-terminal pocket.

Table 5. Complete Hyperparameter Configuration

Hyperparameter	Value
<i>Network Architecture</i>	
Backbone Type	E(3)-Equivariant NodeEdgeNet
Number of Layers (L)	12
Node Hidden Dimension (d)	128
Edge Hidden Dimension	32
Interaction Cutoff Radius	10.0 Å
Residue Connection	Highway Gating
Pre-trained PLM	esm2_t6_8M_UR50D
PLM Embedding Dimension	320 (Layer 6)
Time Embedding Type	Gaussian Smearing (RBF)
Time Embedding Dimension	256
Affinity Fourier Bands (K)	16
Context Dimension	64
<i>Diffusion Process & Loss</i>	
Total Timesteps (T)	1000
Schedule Type	Sigmoid-Based
Scale Start (α_T)	0.9999
Scale End (α_0)	0.0001
Position Loss Weight (λ_{pos})	5.0
Sequence Loss Weight (λ_{seq})	1.0
Edge Type Loss Weight (λ_{edge})	1.0
<i>Optimization</i>	
Optimizer	AdamW
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-6}
Gradient Clipping Norm	50.0
Batch Size	32
Total Training Iterations	500,000
LR Scheduler	ReduceLROnPlateau
Scheduler Patience	20,000 steps

We observe a sharp energy cliff where deviations from the hydrophobic profile result in immediate destabilization. For instance, introducing charged residues like Glutamic Acid (E) or Arginine (R) incurs massive energetic penalties (spiking to positive REU values), caused by severe electrostatic repulsion and desolvation costs within the non-polar groove.

In conclusion, this mutational scan confirms that the model’s high-affinity generations are driven by precise geometric and physicochemical complementarity. The model has effectively learned to navigate the energy landscape, placing residues not just based on sequence probability, but on their ability to minimize the free energy of the structural complex.

D.2. Impact of Guidance Scale

We investigate the behavior of our generative process under varying intensities of Contrastive Affinity Guidance, controlled by the scalar λ . Recall that in our formulation (Eq. 5), λ determines the influence of the *contrastive rectification* term—the gradient component that steers the sampling trajectory away from the abundant low-affinity manifold $p_\theta(\mathbf{Z}|_{s_{\text{low}}})$.

Figure 5 illustrates the trade-off between binding optimization and structural physical validity. We observe two distinct regimes:

Stability-Affinity Equilibrium ($\lambda \in [0.5, 1.0]$): In this lower scaling regime, the model effectively leverages the negative data to prune non-binding modes without compromising structural integrity. The Top-10 candidates achieve their lowest

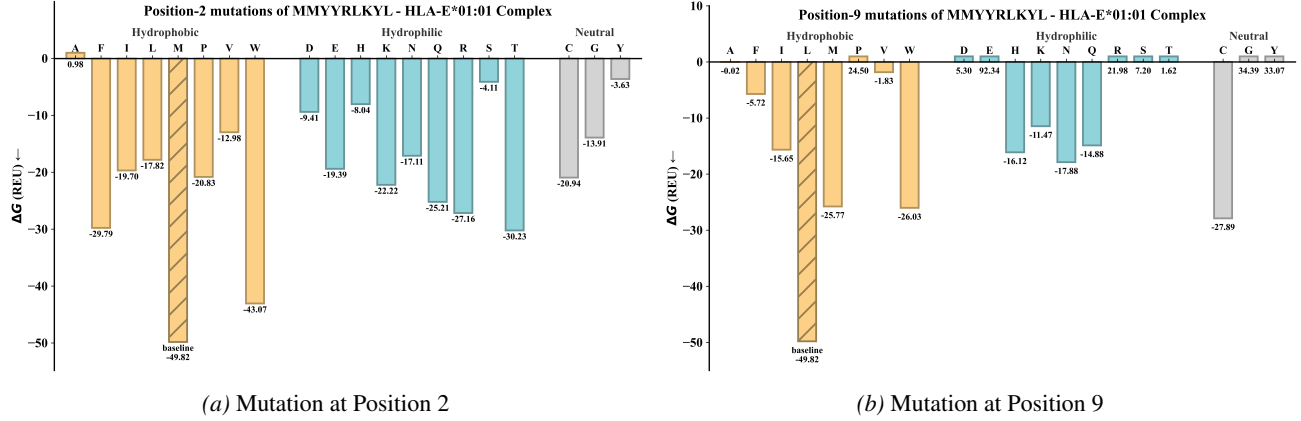


Figure 4. In silico mutational scanning of HLA-E anchor residues. We perform a comprehensive sensitivity analysis on the generated peptide MMYRLKYL by substituting the anchor residues at Position 2 (left) and Position 9 (right) with all 19 alternative amino acids. The bars represent the binding free energy (ΔG) for each variant, categorized by physicochemical properties (Hydrophobic, Hydrophilic, Neutral). The hatched bar represents our model’s generated baseline. The significant energy gap between the baseline and all substitutions confirms that the generated anchors (Met at P2, Leu at P9) correspond to the global energy minimum of the binding pocket.

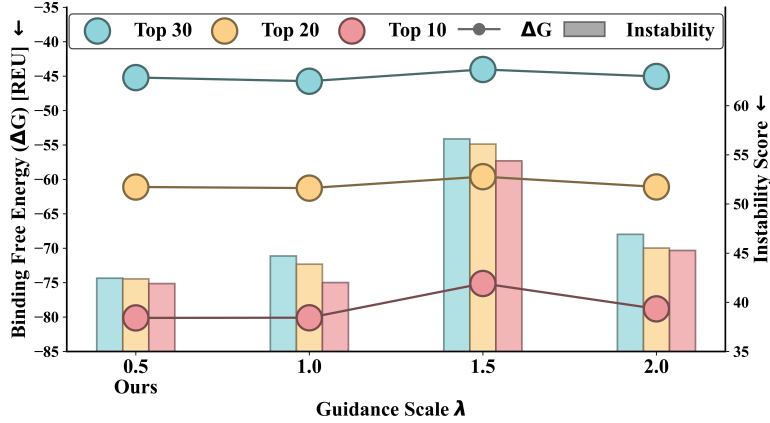


Figure 5. Sensitivity analysis of the Contrastive Affinity Guidance (CAG) scale λ . We report the Binding Free Energy (ΔG [REU], lines, left axis) and Instability Score (bars, right axis) across the Top-10, 20, and 30 generated candidates. Lower values indicate better performance for both metrics. The guidance scale λ modulates the magnitude of the negative gradient rectification term defined in Eq. 5. Results indicate an optimal operating regime where biological stability and binding affinity are simultaneously minimized.

binding free energies ($\Delta G \approx -80$ REU) while maintaining minimal instability scores (≈ 42). This suggests that the directional signal provided by $\nabla \log p_\theta(\mathbf{Z}|s_{\text{low}})$ is sufficiently aligned with the binding landscape to improve affinity, yet gentle enough to keep the generated structures within the chemically plausible region of the prior.

Guidance Saturation ($\lambda \geq 1.5$): As λ increases to 1.5, we observe a sharp degradation in structural quality. The instability score spikes significantly (exceeding 54 for the Top-10 cohort), and the binding energy also worsens. We hypothesize that excessive repulsion from the negative mode forces the Langevin dynamics into high-energy regions of the conformational space that satisfy the contrastive objective but violate geometric constraints. Interestingly, at $\lambda = 2.0$, the model partially recovers in terms of affinity, likely due to the gradient signal dominating the noise, but the instability remains elevated compared to the optimal baseline.

Consequently, we adopt $\lambda = 0.5$ as the default setting for subsequent experiments, as it maximizes binding affinity while preserving the stability required for viable binders.